USEI08 - Search by Geographical Area: Complexity Analysis

Context

Geographic range search enables efficient retrieval of all train stations within a rectangular bounding box defined by latitude and longitude boundaries. The 2D KD-Tree structure provides spatial partitioning that allows pruning of irrelevant subtrees, avoiding the need to scan the entire dataset sequentially.

User Story

As a planner, I want to query the 2D-tree for all stations in a latitude/longitude range with optional filters, so I can extract relevant stations quickly.

Acceptance Criteria

The region: lat [latMin, latMax], lon [lonMin, lonMax] (inclusive).

The search should allow the optional application of filters with the following criteria: - isCity (true | false) - isMainStation (true | false) - country (PT | ES | all)

Returns: 3-5 sample queries; Temporal analysis complexity.

Temporal Complexity Analysis

1. Range Search Algorithm

The search is performed recursively on the 2D KD-Tree:

1. **Check if current node's coordinates fall within the region**
2. **Apply filter criteria** to stations at matching nodes
3. **Determine which subtrees to explore** based on pruning conditions
4. **Recurse on left and/or right subtrees** depending on the splitting axis

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Detailed Complexity Analysis (Code Excerpt)

```
private void rangeSearchWithFilterRecursive(Node node,
                                            double minLat, double maxLat,
                                            double minLon, double maxLon,
                                            StationFilter filter,
                                            int depth,
                                            List<Station> result) {
    if (node == null) return;  // O(1)
```

```
    // Check if node is in range - O(1)
    if (node.latitude >= minLat && node.latitude <= maxLat &&
        node.longitude >= minLon && node.longitude <= maxLon) {

        // Add filtered stations - O(b) where b = bucket size
        for (Station station : node.stations) {
            if (filter == null || filter.matches(station)) {  // O(1)
                result.add(station);  // O(1) amortized
            }
        }
    }

    boolean splitByLatitude = (depth % 2 == 0);

    // PRUNING LOGIC - O(1)
    if (splitByLatitude) {
        if (minLat <= node.latitude) {
            rangeSearchWithFilterRecursive(node.left, ...);  // T(n/2)
        }
        if (maxLat >= node.latitude) {
            rangeSearchWithFilterRecursive(node.right, ...); // T(n/2)
        }
    } else {
        if (minLon <= node.longitude) {
            rangeSearchWithFilterRecursive(node.left, ...);  // T(n/2)
        }
        if (maxLon >= node.longitude) {
            rangeSearchWithFilterRecursive(node.right, ...); // T(n/2)
        }
    }
}
```

Per-Call Complexity Breakdown

Operation	Complexity	Description
Null check	O(1)	Base case validation
Coordinate comparison	O(1)	4 boundary checks (lat/lon min/max)
Bucket iteration	O(b)	Process b stations at same coordinates
Filter matching	O(1)	3 boolean/string comparisons per station
Result collection	O(1)	ArrayList.add() amortized
Pruning decision	O(1)	Determine which subtrees to explore
Per-node total	**O(b)**	Typically b 3

Overall Complexity Analysis

Best Case: O(log n + k) **When:** Small region, sparse area, highly effective pruning

Why: - Only nodes along the path to the target region are visited - Most subtrees are pruned without traversal - Tree height is log n due to balanced construction - Minimal intersection with splitting planes

Recurrence:

```
T(n) = T(n/2) + O(1)  →  T(n) = O(log n)
Total: O(log n + k) where k = results found
```

Example: - Region: 0.1° × 0.1° (small urban area) - Dataset: 64,000 stations - Nodes visited: ~20-30 - Time: ~5-10ms

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Average Case: O(√n + k) **When:** Medium-sized region (1-5 degrees), typical geographic query

Why: - A rectangular range query intersects approximately √n splitting planes in a 2D KD-Tree - This is a well-established property for orthogonal range queries in k-dimensional trees - The query region spans both sides of many splits, preventing complete pruning - Alternating latitude/longitude splits at each level

Mathematical Reasoning:

```
For a balanced 2D-tree with n nodes:
- Number of splitting hyperplanes intersected  O(√n)
- Each intersected plane requires visiting both subtrees
- Total nodes visited = O(√n)
```

Example: - Region: Portugal (3° × 2°) - Dataset: 64,000 stations - Nodes visited: ~250-300 - Time: ~15-30ms

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Worst Case: O(n + k) **When:** Region covers entire dataset, no effective pruning

Why: - Bounding box encompasses all or most of the coordinate space - Pruning conditions are always satisfied: `minLat  node.latitude  →  always true` `maxLat  node.latitude  →  always true` - Every node in the tree must be visited - Degenerates to complete tree traversal

Specific Scenarios: 1. **Full map coverage:** Europe-wide query 2. **Wide horizontal band:** Latitude range covers 30+ degrees 3. **Wide vertical band:** Longitude range covers 40+ degrees

Example: - Region: All Europe (latMin=35, latMax=71, lonMin=-10, lonMax=40) - Dataset: 64,000 stations - Nodes visited: ~64,000 (all) - Time: ~100-150ms

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2. Filter Application Complexity

The `StationFilter.matches()` method has **O(1)** complexity:

```
public boolean matches(Station station) {
    // Check isCity filter - O(1)
    if (isCity != null && station.isCity() != isCity) {
        return false;
    }

    // Check isMainStation filter - O(1)
    if (isMainStation != null && station.isMainStation() != isMainStation) {
        return false;
    }

    // Check country filter - O(1)
    // (country codes are 2-3 chars, equalsIgnoreCase is O(m) but m is constant)
    if (country != null && !country.equalsIgnoreCase(station.getCountry())) {
        return false;
    }

    return true;
}
```

Filter Impact: - Applied **only** to stations already in the geographic region - Does **not affect** tree traversal complexity - Total filter cost: O(k) where k = stations in region

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3. Region Validation Complexity

```
public GeographicRegion(double latMin, double latMax, double lonMin, double lonMax) {
    validateLatitude(latMin, "latMin");     // O(1)
    validateLatitude(latMax, "latMax");     // O(1)
    validateLongitude(lonMin, "lonMin");    // O(1)
    validateLongitude(lonMax, "lonMax");    // O(1)

    // Range validation - O(1)
```

```
    if (latMin > latMax) { /* ... */ }
    if (lonMin > lonMax) { /* ... */ }

    // Assignment - O(1)
}
```

Total: O(1)

Summary Table

Operation	Best Case	Average Case	Worst Case
Range Search	O(log n + k)	O(√n + k)	O(n + k)
Region Validation	O(1)	O(1)	O(1)
Filter Application	O(k)	O(k)	O(k)
Result Collection	O(k)	O(k)	O(k)

Where: - n = total nodes in tree (~64,000) - k = number of matching stations - b = bucket size (typically 1-3)

Pruning Effectiveness by Region Size

Region Size	Area Coverage	Pruning Rate	Nodes Visited	Complexity
Very Small (< 0.5°)	< 1%	~95%	O(log n)	O(log n + k)
Small (0.5° - 2°)	1-5%	~80%	O(√n)	O(√n + k)
Medium (2° - 5°)	5-20%	~60%	O(√n)	O(√n + k)
Large (5° - 15°)	20-50%	~30%	O(n/2)	O(n + k)
Very Large (> 15°)	> 50%	< 10%	O(n)	O(n + k)

Comparison with Linear Scan

Approach	Complexity	Typical Time (64k stations)	Speedup
Linear Scan	O(n)	50-100ms	1x
2D-Tree (Small)	O(log n + k)	5-10ms	**10x**
2D-Tree (Medium)	O(√n + k)	15-30ms	**3-5x**
2D-Tree (Large)	O(n + k)	100-150ms	1x

Key Insight: The 2D-Tree provides significant speedup for localized queries, which represent the majority of real-world use cases.

Empirical Measurement

The implementation tracks execution time using nanosecond precision:

```
public List<Station> rangeSearchWithFilter(GeographicRegion region,
                                           StationFilter filter) {
    long startTime = System.nanoTime();

    rangeSearchWithFilterRecursive(...);

    long endTime = System.nanoTime();
    this.lastQueryTimeNanos = endTime - startTime;

    return result;
}
```

Access via: - `getLastQueryTimeNanos()` → nanoseconds - `getLastQueryTimeMs()` → milliseconds - `SearchResult.getExecutionTimeMs()` → included in result object

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Conclusion

The 2D KD-Tree range search achieves $\mathbf{O(\sqrt{n} + k)}$ average-case performance through spatial partitioning and effective pruning. This represents a substantial improvement over O(n) linear scanning for typical geographic queries, with the square-root factor arising naturally from the geometry of rectangular range searches in two-dimensional space. The implementation's balanced tree construction ensures predictable performance, while optional filtering adds negligible overhead.